

Molecular Simulations of Thermosetting Polymers

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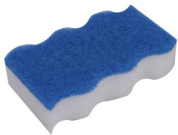


cameronabrams.github.io/thermosets-2026



Thermosetting Polymers: Extraordinary Everyday Materials

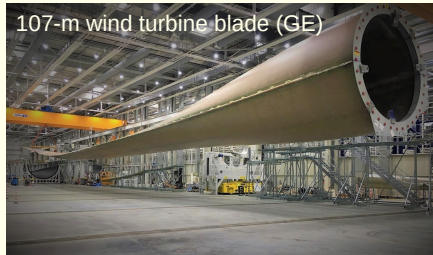
Melamine “sponge”



Solid-fuel ramjet missile
(Norway)



107-m wind turbine blade (GE)



Adhesives



Vintage bakelite billiard balls



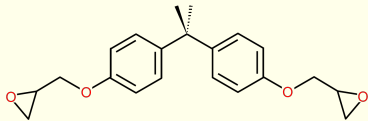
Boeing 787 Dreamliner (Wikipedia)



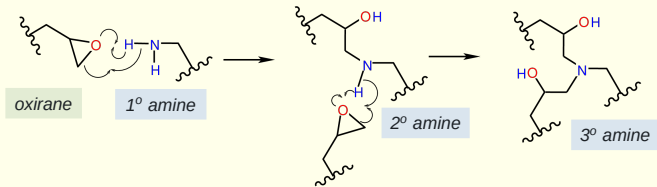
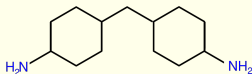
Common Thermosets: Epoxy Resins and Amine Hardeners

Examples:

Diglycidyl ether of bisphenol A (DGEBA)



4,4'-diaminodicyclohexylmethane (PACM)



Thermosetting Polymer Design and Use Considerations

Strong and light

Easily processed
*Low-viscosity
prepolymer*

Hard to recycle

strength

toughness

Corrosion/
chemical
resistant

Insulating

Rely on
petrochemical
precursors

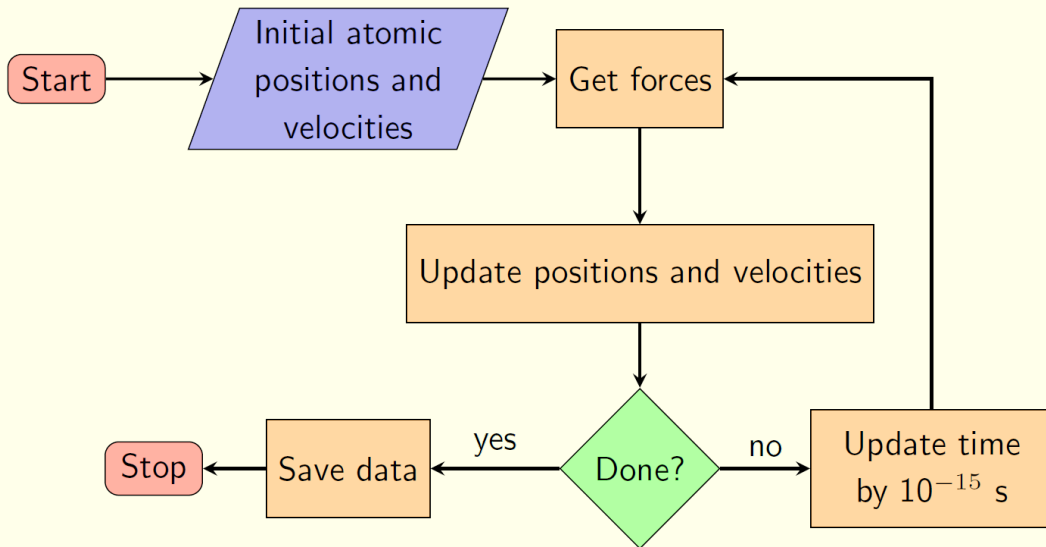
thermomechanical
stability

toughness

A key imperative: Improve *toughness* without sacrificing strength and thermomechanical stability

What generic topological features determine thermoset toughness, and how can we realize them in actual systems?

Molecular Dynamics (MD) Simulations

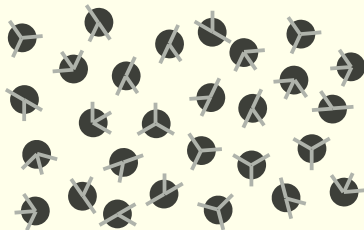
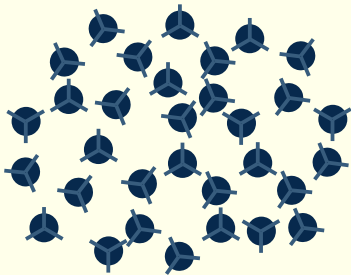


Early work: Generic Thermosets and 3D Network Regularity

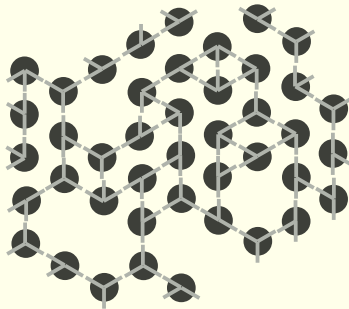
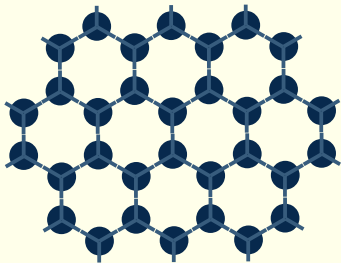
Uniform bond valence angles; $z = 3$

Nonuniform bond valence angles ; $z = 3$

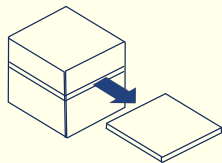
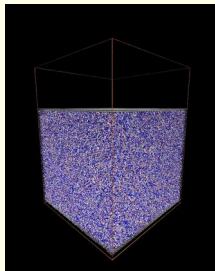
Liquid



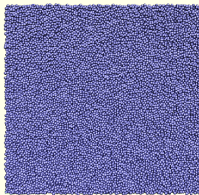
Network
polymer



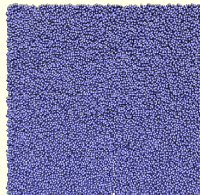
Protovoid-based strain hardening of generic network polymers



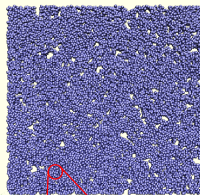
510,000 LJ particles



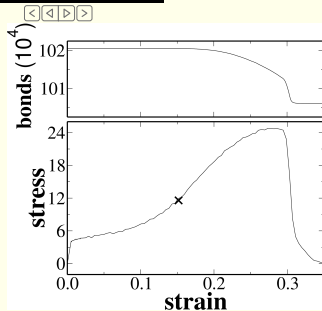
liquid



95% crosslinked
 $z=4$

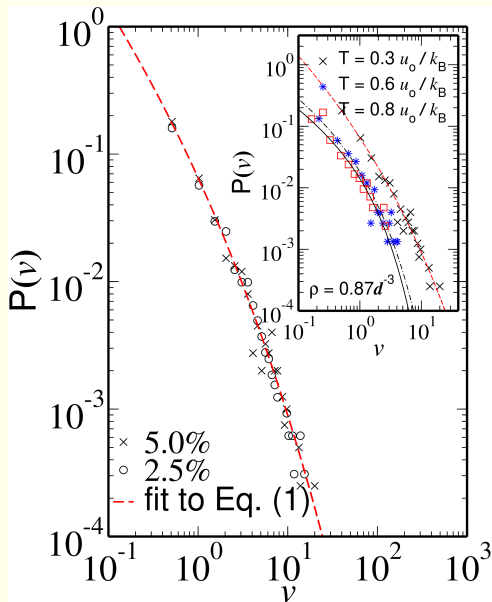


5% tensile strain
 $\dot{\epsilon} = 2 \times 10^{-4} \tau^{-1}$



Voids open before bonds start breaking

Void sizes conform to equilibrium free-volume theory



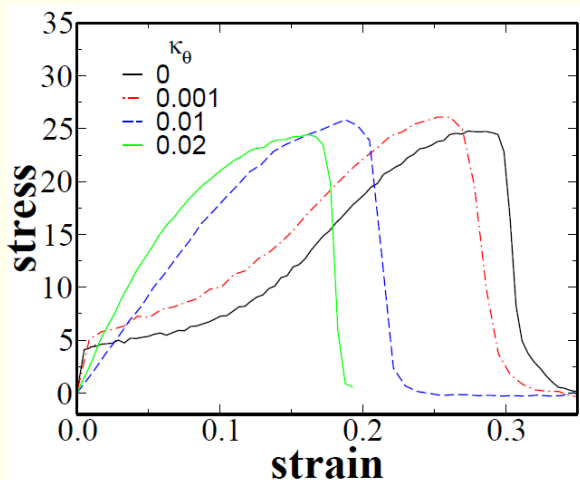
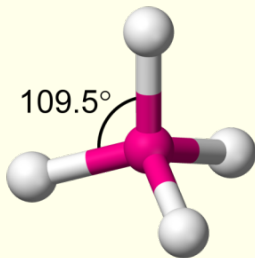
$$P(v) \propto v^{\alpha-1} \exp\left(-\frac{v^\zeta}{k_B T}\right)$$

Hoover, Hoover & Hanson. J Chem Phys 1979;70:1837

- Protovoids are encoded by bonding geometry
- Protovoids are a “natural” part of a system’s void spectrum
- Strain provides energy to actualize protovoids

Voids can be suppressed using constrained bond angles

$$V_{\text{angle}}(\theta_{ijk}) = \kappa_{\theta} (\theta_{ijk} - \theta_0)^2$$

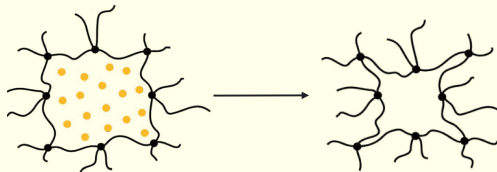


Protovoid: a definition

- A void encoded by bonding geometry *before* any strain is applied
- Latent free volume that strain converts into real void space
- Suppressed by constraining bond angles — therefore designable

Engineering protovoid-based toughening into real epoxy thermosets

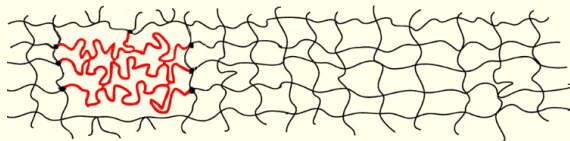
Reactive encapsulation of solvent and drying (RESO)



Sharifi, Jang, Abrams, Palmese. J Mater Chem A 2014;2:16071

Jang, Sharifi, Palmese, Abrams. Polymer 2014;55:3859

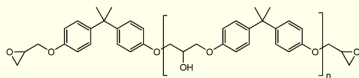
Partially reacted substructures (PRS)



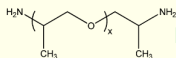
Sharifi, Jang, Abrams, Palmese. Macromolecules 2015;48:7495

Jang, Sharifi, Palmese, Abrams. Polymer 2016;90:249

Reactive encapsulation of solvent and drying (RESO): Experiments



Diglycidyl Ether of Bisphenol-A (DGEBA), EPON 828, $n=0.13$



Polyetheramine (Jeffamine D400), $x=6.1$

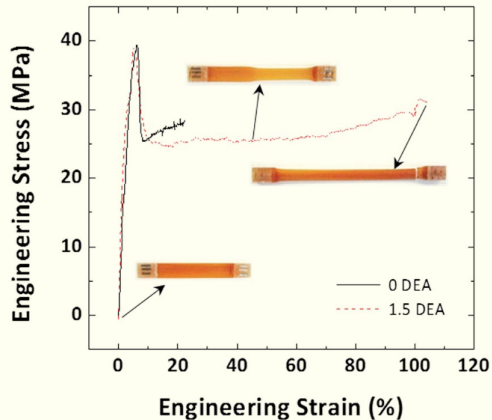
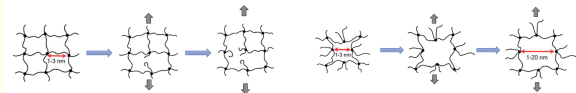
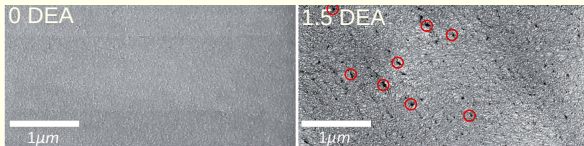
Low T_g



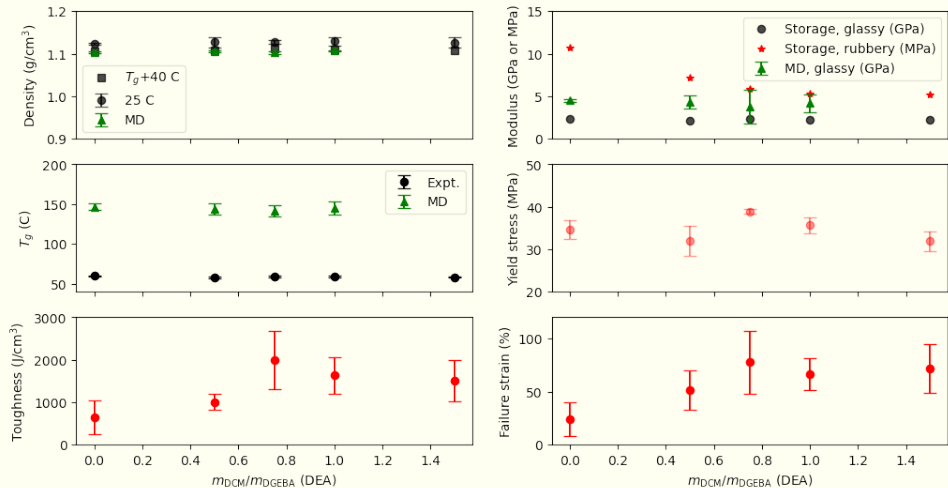
Dichloromethane (DCM)

$$DEA = \frac{m_{DCM}}{m_{DGEBA}}$$

Fracture surfaces after tensile failure



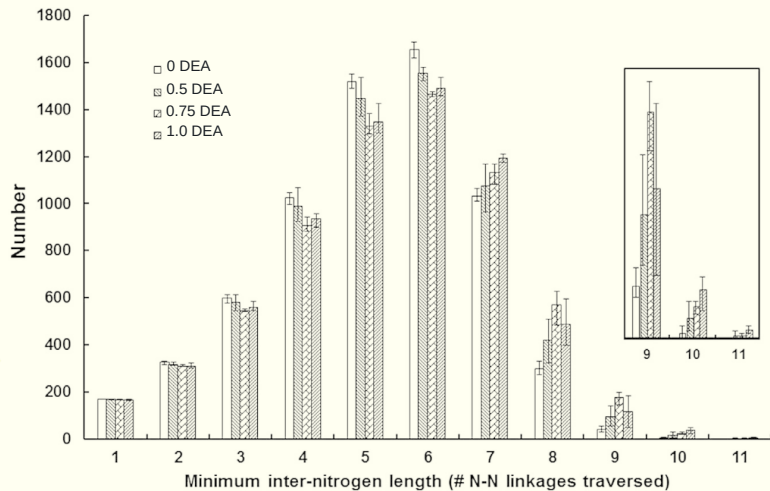
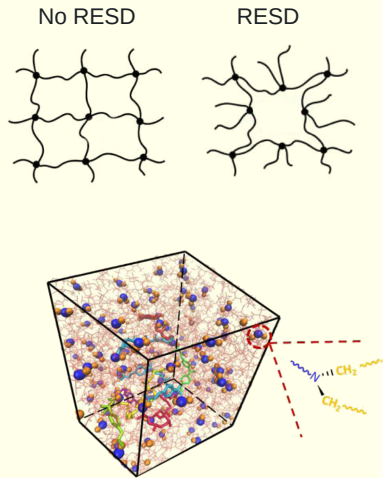
RESO: Solvent effects partially reproduced by MD



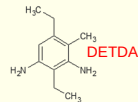
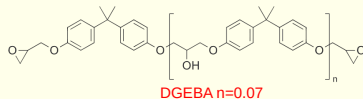
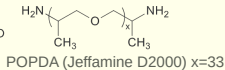
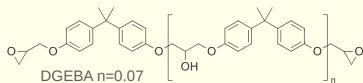
Sharifi, Jang, Abrams, Palmese. J Mater Chem A 2014;2:16071

Jang, Sharifi, Palmese, Abrams. Polymer 2014;55:3859

MD-simulated RESD network architecture is consistent with voids

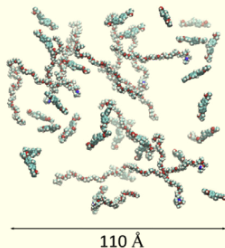


Partially reacted substructures (PRS)

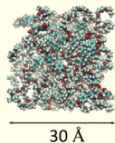


Partial curing of low- T_g network into a PRS

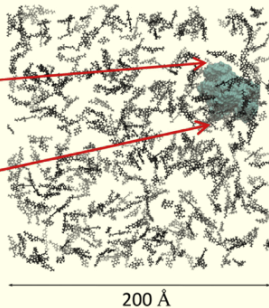
Curing of **high- T_g** network matrix with embedded PRS



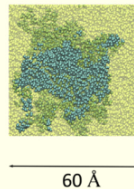
500 ps



Packing



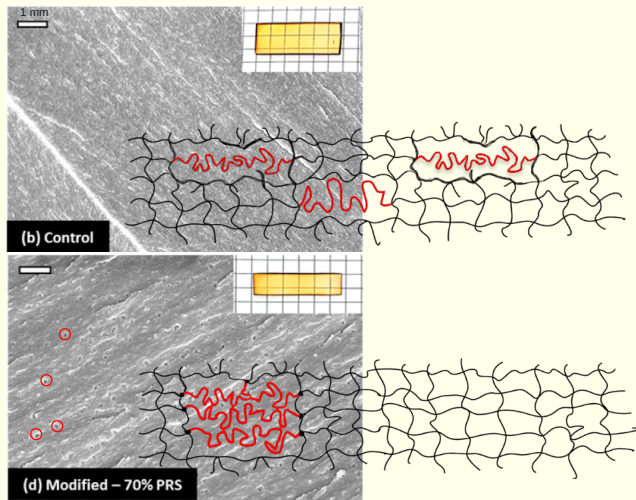
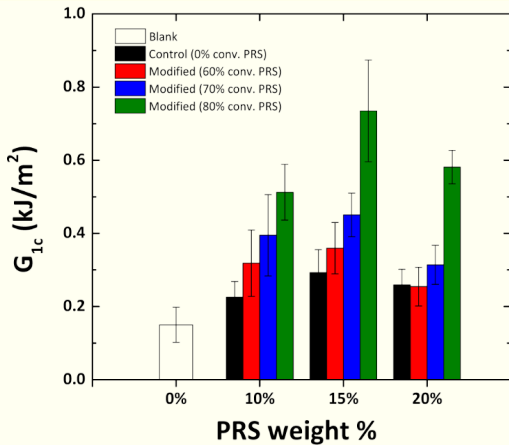
Cured



Variables: (1) Degree of PRS cure and (2) wt % of PRS in composite system

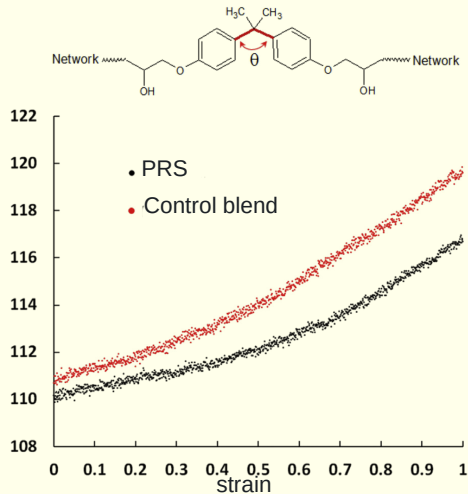
PRS: Protovoid-induced toughness enhancement

G_{1c} : strain energy release rate

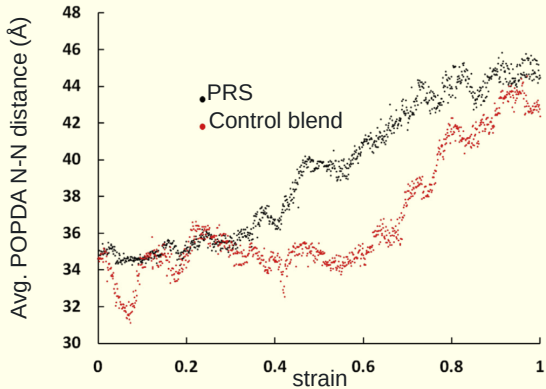


PRS: MD simulations of 15-wt% of 60% cured PRS vs. control blend

Avg. DGEBA central C-C-C angle (degrees)

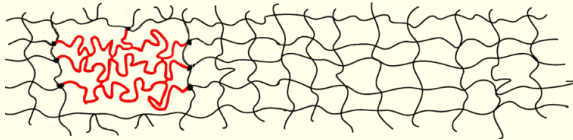
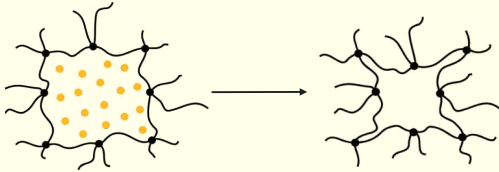


Angle-loading is delayed in PRS



PRS yields as POPDA's stretch cooperatively

The story so far...



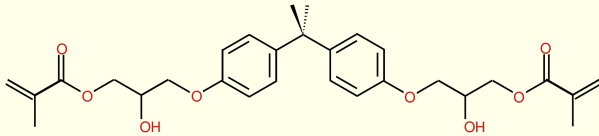
Control of network architecture is key for improving toughness without sacrificing modulus and thermal stability (T_g)

So far, only specialized blends have been used...

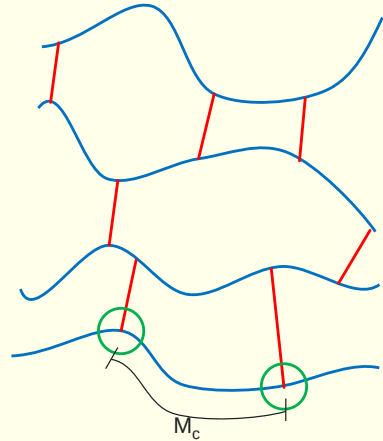
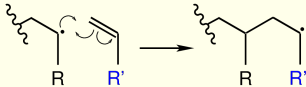
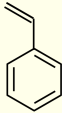
Common Thermosets: Vinyl Ester Resins and Styrene Diluents

Examples:

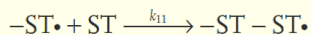
Bisphenol A-glycidyl methacrylate (bis-GMA) "VE"



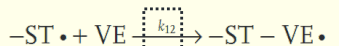
Styrene "ST"



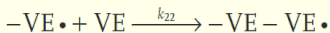
Reactivity ratios in vinyl-ester/styrene thermosets



$$k \propto p$$



Assumed equal



$$r_1 = \frac{k_{11}}{k_{12}} = \frac{p_{11}}{p_{12}}$$

$$p_{11} = r_1 p_{12}$$

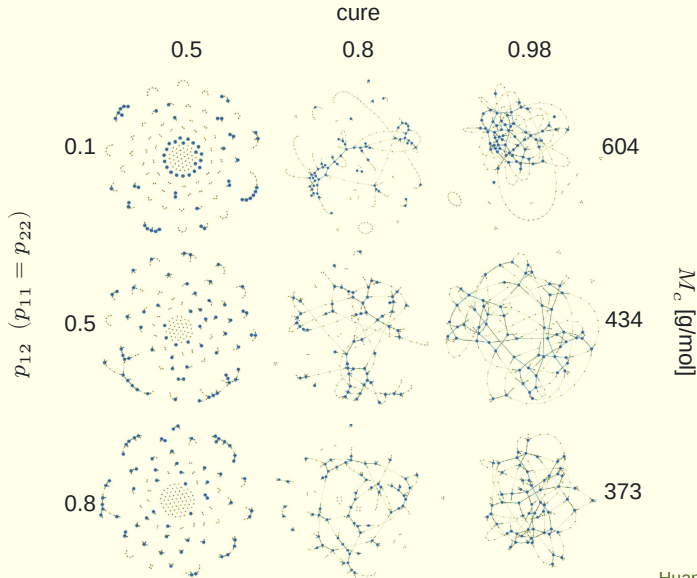
$$p_{22} = r_2 p_{12}$$

$$r_2 = \frac{k_{22}}{k_{21}} = \frac{p_{22}}{p_{21}}$$

$$p_{11} + p_{22} + p_{12} = 1$$

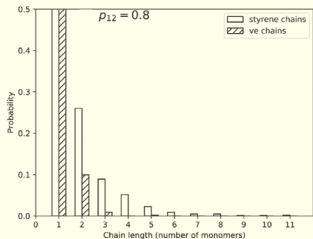
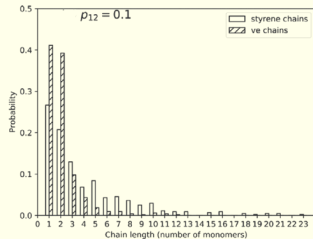
Temperature [°C]	r_1	r_2	p_{11}	p_{22}	p_{12}
30 ^{a)}	0.45	1.54	0.15	0.52	0.33
90 ^{a)}	0.47	0.61	0.23	0.29	0.48
90 ^{b)}	0.49	0.5	0.25	0.25	0.50
140 ^{c)}	0.36	0.24	0.23	0.14	0.63

Network topology: sensitivity to p_{12} at constant VE:ST



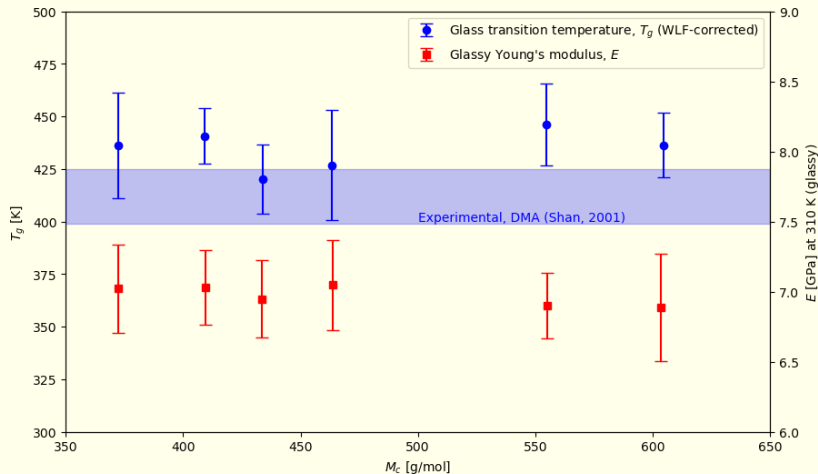
- Higher p_{12} means more VE-ST cross-propagation.
- M_c falls from 604 to 373 g/mol at fixed VE:ST.
- Composition alone does not set topology.

Chain-length distributions shift sharply with p_{12}



- At $p_{12} = 0.1$, styrene homopropagates into long chains — a distribution with a heavy tail out past 20 monomers.
- At $p_{12} = 0.8$, cross-propagation truncates everything: almost nothing exceeds 5 monomers.
- Same VE:ST ratio in both panels. Only the reactivity ratio changed.

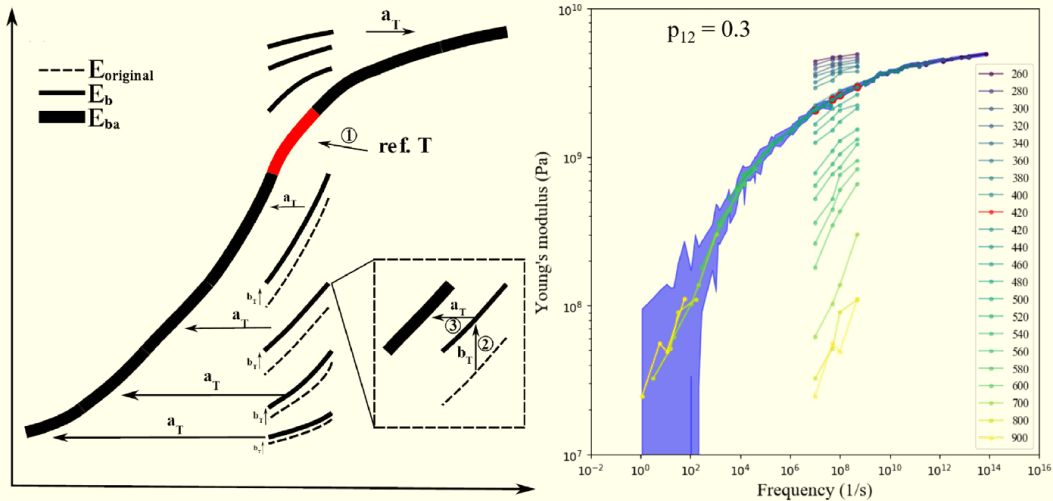
Network topology: insensitivity of T_g and E at constant VE:ST



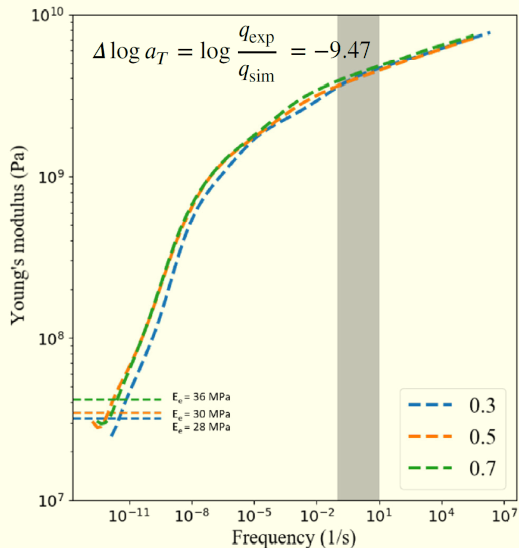
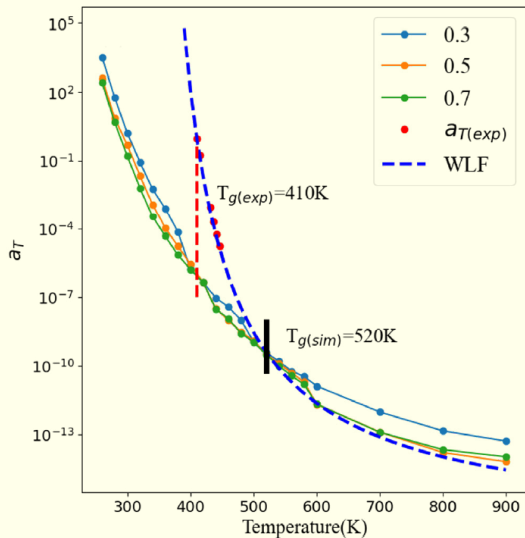
WLF corrects T_g for heating-rate discrepancy

Young's modulus is a factor of 2 too high!

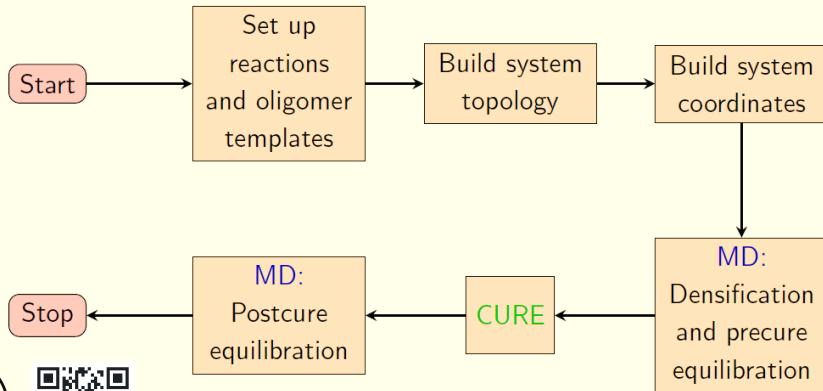
Estimating quasi-static E using time-temperature superposition



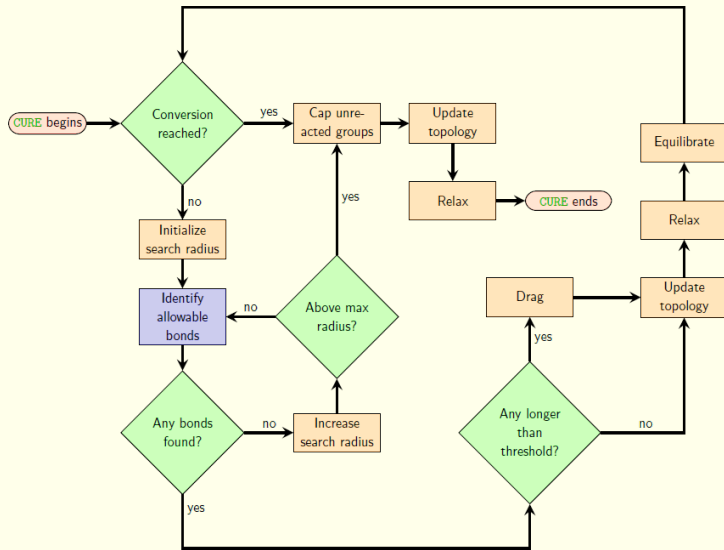
TTSP with cooling-rate correction gives correct glassy modulus



htpolynet: Network Polymer MD System Generator



htpolynet: The CURE Algorithm



htpolynet example: DGEBA/PACM (inputs)

User input: one YAML file



DGEBA

DGE:

```
smiles: "CC(C)(c1ccc(OC[CH:3]([OH:5])[CH3:1])cc1)c1ccc(OC[CH:4]([OH:6])[CH3:2])cc1"
reactive_atoms: {1: C1, 2: C2, 3: C3, 4: C4, 5: O1, 6: O2}
count: 200
```



PACM

PAC:

```
smiles: "C1C[CH:1](CCC1CC2CC[CH:2](CC2)[NH2:3])[NH2:4]"
reactive_atoms: {1: C1, 2: C1, 3: N1, 4: N1}
count: 100
```



Chemistry
+ config.

reactions:

- name: Primary-to-secondary-amine
reactants: {1: PAC, 2: DGE}
product: PAC~N1-C1-DGE
- name: Secondary-to-tertiary-amine
reactants: {1: PAC~N1-C1-DGE, 2: DGE}
product: PAC~N1-C1-DGE-C1-DGE

CURE:

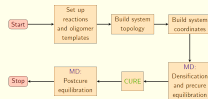
controls:

```
search_radius: 0.5 # nm
desired_conversion: 0.95
```

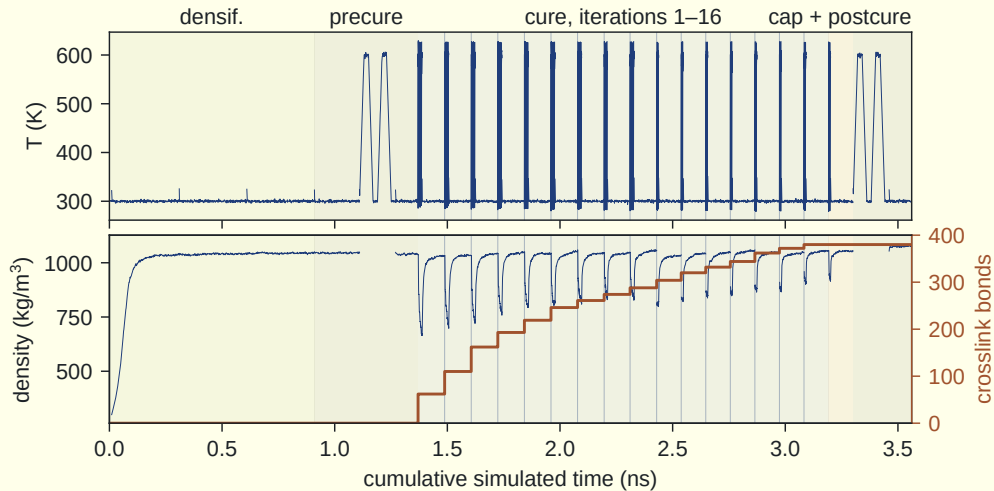
densification: ...

precure: ... postcure: ...

Monomers are built from atom-mapped SMILES (the map numbers mark reactive atoms).
htpolynet generates every oligomer template it needs from the two reactions and their symmetry equivalents, parameterizes them with GAFF, then densifies, cures and equilibrates.

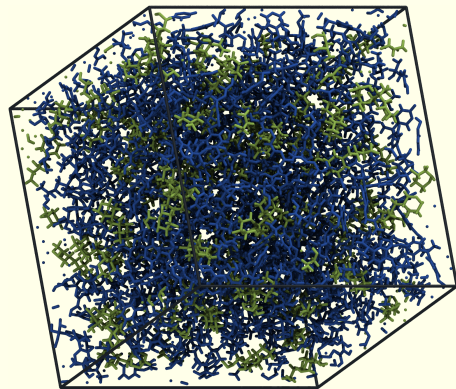
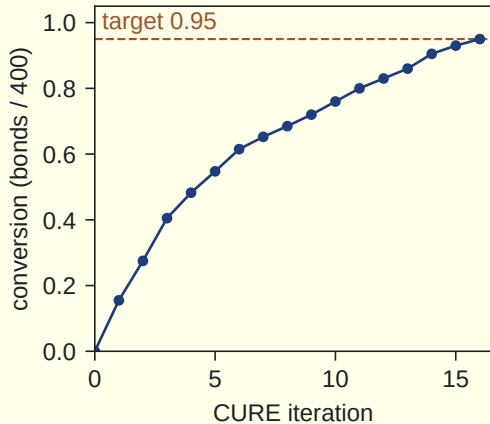


htpolynet output: the build, stage by stage



Example 3: 200 DGEBA + 100 PACM, htpolynet 2.8. 16 cure iterations reach 380 of 400 possible crosslinks (0.95). The 600 K spikes are the relax stages inside each iteration.

htpolynet output: cure conversion and the final network

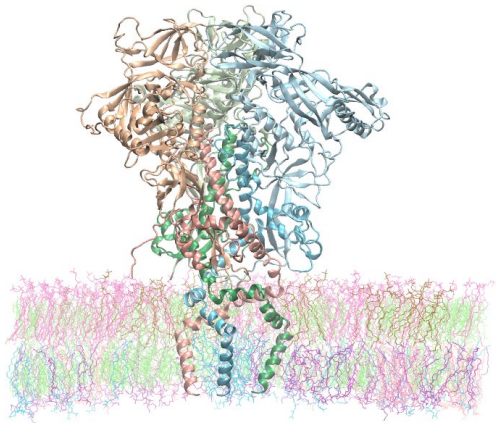


box edge 5.14 nm

■ DGEBA ■ PACM

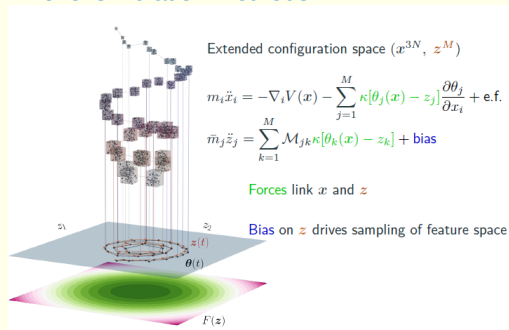
Final density at 300 K: 1076 kg/m³ (100 ps NPT average).
One network of 13,802 atoms, plus 2 unreacted DGEBA.

Modeling the pretriggered HIV-1 spike

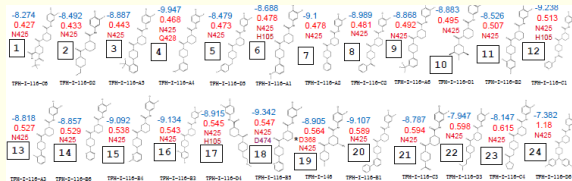


(built using pestifer [<https://pestifer.readthedocs.io/en/latest/>])

Novel simulation methods: TAMMD



Computational Drug Design



Key takeaways

- Thermosets can be toughened without sacrificing thermomechanics via *protovoids*
- Protovoid-based toughening demonstrated in generic MD simulations and several experimental systems
- Protovoids are a class of generic topological features that can be leveraged to implement new toughening approaches
- Network topology can be controlled independently of monomer composition via reactivity ratios

Acknowledgments

Those who did the work discussed in this talk:

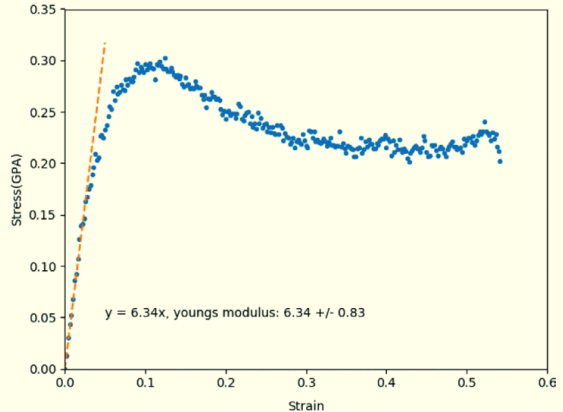
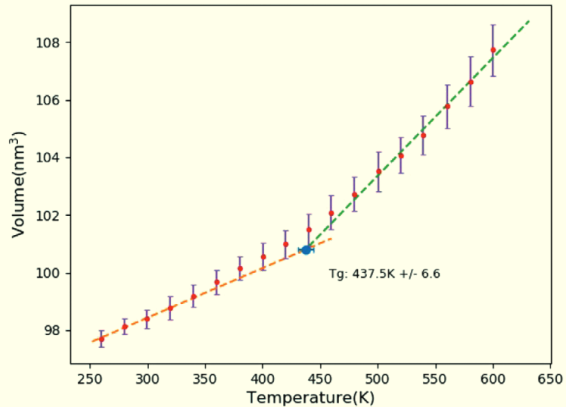
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Funding for the work discussed in this talk:

- US Army W911NF-06-2-0013, W911NF-07-1-0301, W911NF-12-2-0022, W911NF-12-R-0011, W911NF-13-2-0046, W911NF-17-2-0227
- NSF AST-0959884
- ACS PRF 42368-G7

Backup slides

Examples of measuring T_g and E (glassy)



WLF-correction to T_g

$$\Delta T_g = \frac{-C_2 \log_{10} \frac{q_{\text{exp}}}{q_{\text{sim}}}}{C_1 + \log_{10} \frac{q_{\text{exp}}}{q_{\text{sim}}}} \quad (C_1 = 17.3, C_2 = 92.4) \text{ with } q_{\text{exp}} = 3.33 \text{ K/s and } q_{\text{sim}} = 10^{10} \text{ K/s}$$